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B.Tech. Degree III Semester Examination November 2014

ME 1304 FLUID MECHANICS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) Define and explain the Newton's law of viscosity.
 (b) What are the conditions of equilibrium of a floating body and a submerged body?
 (c) What is a pitot-tube? How will you determine the velocity at any point with the help of pitot tube?
 (d) Explain minor losses in pipes.
 (e) Define the terms: (i) velocity potential function (ii) stream function
 (f) What is magnus effect? Explain.
 (g) What do you understand by the terms boundary layer and boundary layer thickness?
 (h) What do you mean by Karman vortex street? Explain with a neat diagram.

PART B

(4 x 15 = 60)

- II. (a) The pressure intensity at a point in a fluid is given 4.9 N/cm^2 . Find the corresponding height of fluid when it is (i) water (ii) an oil of specific gravity 0.8. (7)
 (b) Calculate the capillary effect in millimeters in a glass tube of 4 mm diameter, when immersed in (i) water (ii) mercury. The values of the surface tension of water and mercury in contact with air are 0.075 N/m and 0.52 N/m respectively. The angle of contact for water is zero that for mercury 130° . Take density of water is 1000 kg/m^3 . (8)
- OR**
- III. (a) A circular plate 3 m diameter is immersed in water in such a way that its greatest and least depth below the free surface are 4 m and 1.5 m respectively. Determine the total pressure on one face of the plate and position of centre of pressure. (7)
 (b) What is the difference between U tube differential manometers and inverted U tube differential manometers? Where are they used? (8)
- IV. (a) What is Euler's equation of motion? How will you obtain Bernoulli's equation from it? (8)
 (b) Derive Darcy-Weisbach equation. (7)
- OR**
- V. (a) An oil of specific gravity 0.8 is flowing through a venturimeter having inlet diameter 20 cm and throat diameter 10 cm. The oil-mercury differential manometer shows a reading of 25 cm. Calculate the discharge of oil through the horizontal venturimeter. Take $C_d = 0.98$. (7)
 (b) Write short notes on the following: (8)
 (i) Laminar flow
 (ii) Turbulent flow
 (iii) Dimensions of flow

(P.T.O.)

- VI. (a) The velocity potential function (ϕ) is given by an expression (7)

$$\phi = \frac{-xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$$

- (i) Find the velocity components in x and y directions.
 (ii) Show that ϕ represents a possible case of flow.

- (b) Write short notes on the following: (8)

- (i) Vortex tube
 (ii) Doublet
 (iii) Irrotational flow
 (iv) Vorticity

OR

- VII. (a) The velocity vector in a fluid flow is given $V = 4x^3 i - 10x^2y j + 2t k$ (8)

Find the velocity and acceleration of a fluid particle at (2, 1, 3) at time $t = 1$.

- (b) Write short notes on the following: (7)

- (i) Convective acceleration
 (ii) Drag and Lift
 (iii) Equation of streamline

- VIII. Derive Prandtl's boundary layer equations. (15)

OR

- IX. Write short notes on the following: (15)

- (i) Skin friction drag
 (ii) Momentum integral equation
